

MapReduce

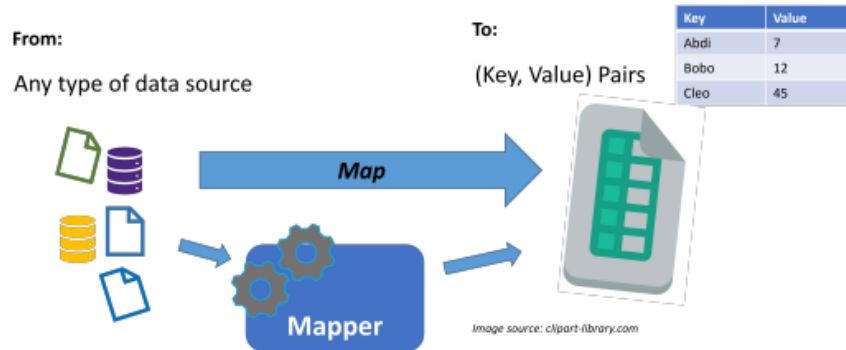


What is MapReduce?

- A programming paradigm
- Use cases
 - Distributed computing
 - BigData analysis in a distributed system based on Java
 - Write application which process data in parallel
- Two main functions
 - **Map:**
 - Filters and sorts data, converts data into (key, value) pairs
 - **Reduce:**
 - Take the input from **Map** and **reduces** its size by some operations

Map function

- Takes a set of data and converts it into another set of data, where individual elements are broken down into tuples (**key-value**) pairs.
- Executed in **multiple machines** at the **same time** in a distributed system



Key-Value Pair

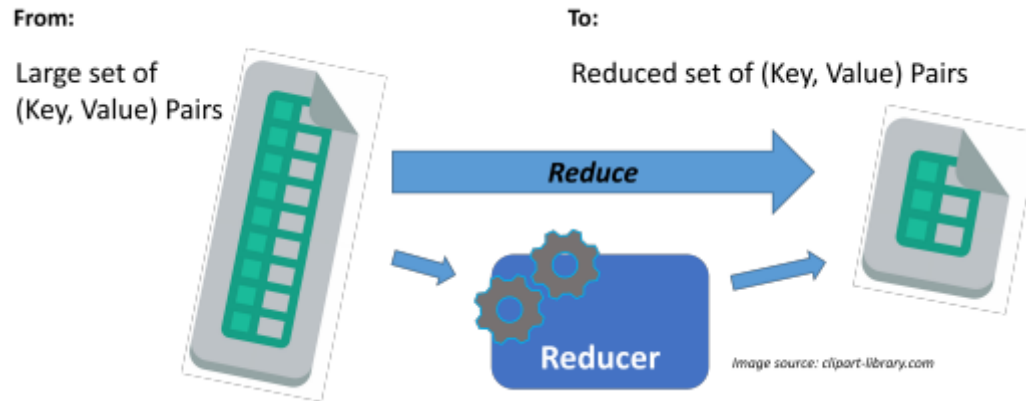
- Look up by “key” => Get a “value”
- Real-life examples:
 - Dictionary: Look up by a “word” => Get an “explanation”
 - Phone directory: Look up by a “Name” => Get a “Phone number”

Key	Value
Firewall	A digital guard protecting networks
Cloud	A place where data lives, not rain
Gravity	The reason things fall down
Keyboard	A tool for talking to computers
Pixel	The smallest unit of a digital image

- Key must be immutable (unchangeable)

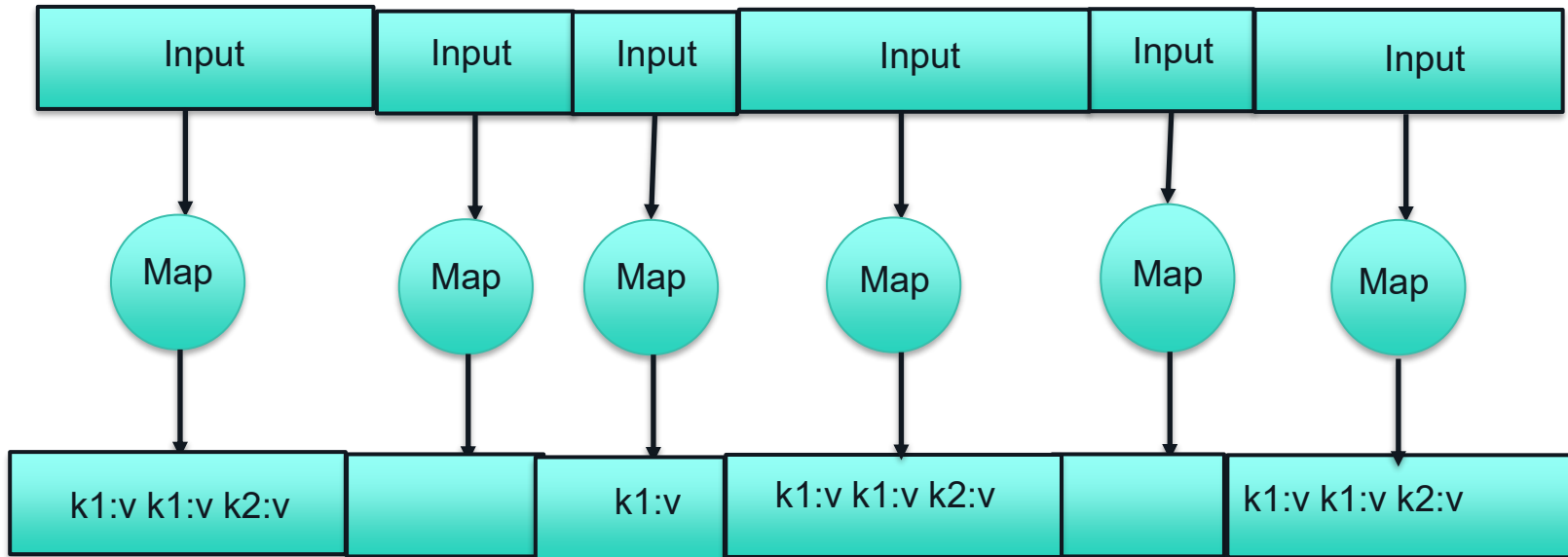
Reduce function

- Combines data tuples (key-value pairs) into a smaller set of tuples.



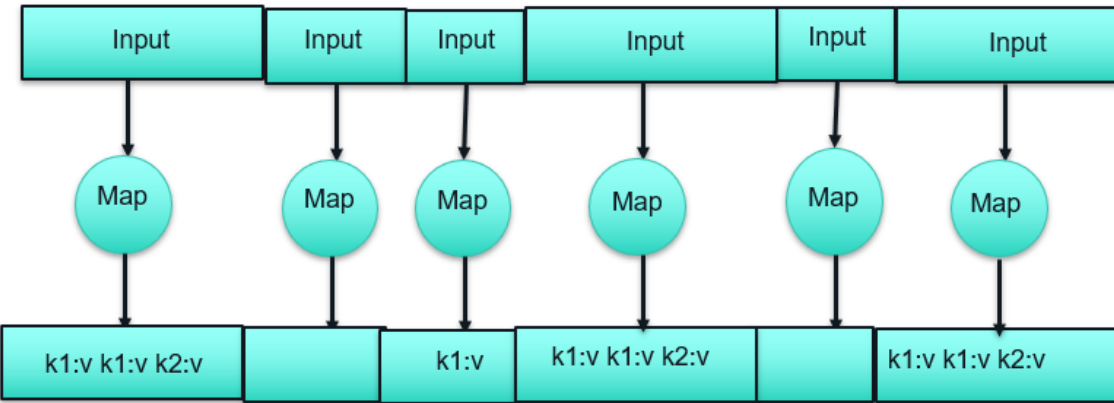
Map

➤ (Key, value)



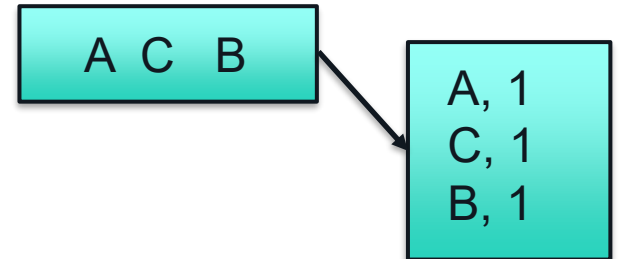
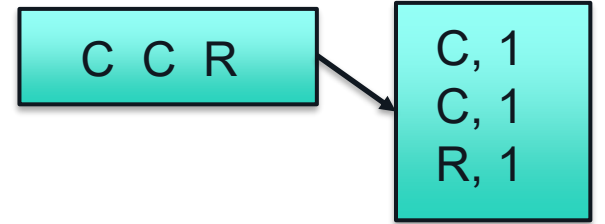
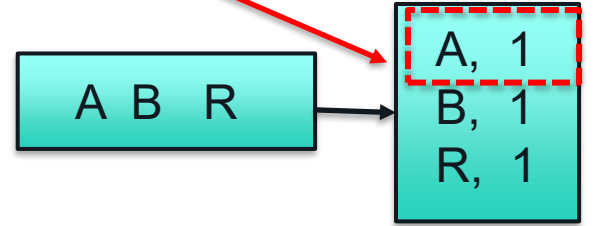
(Key, value) pair

Example of (Key, value) pairs



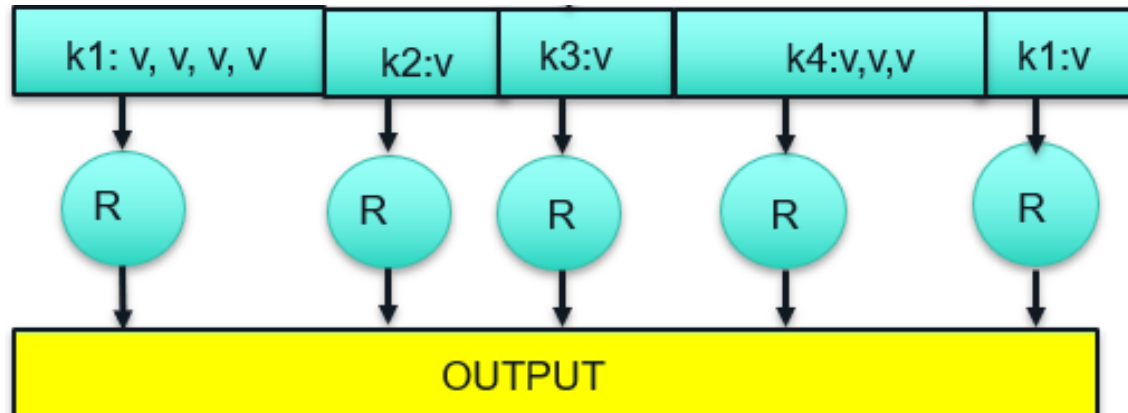
A, 1

A: key
1: value

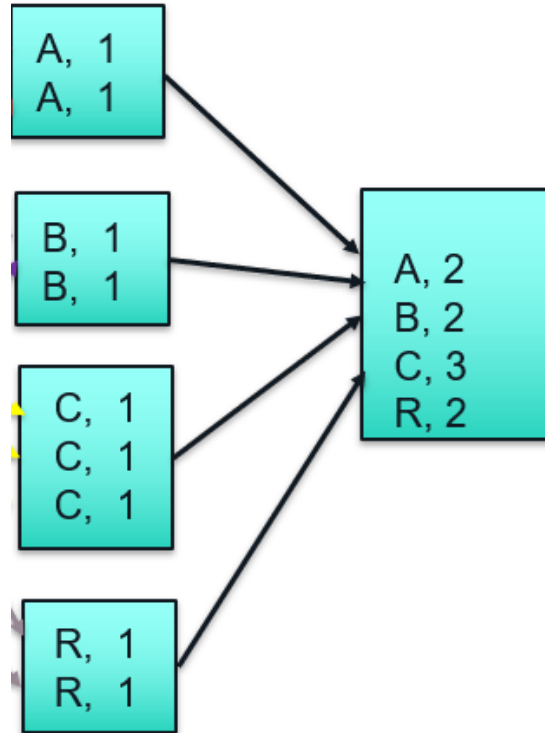


Reduce function

- The Reduce function takes the output from Map function as its input and then reduces its size by performing some kinds of termination operations on the dataset



Example of Reduce function



Map + Reduce = MapReduce

From:

Any type of data source

To:

Reduced set of Key-Value Pairs

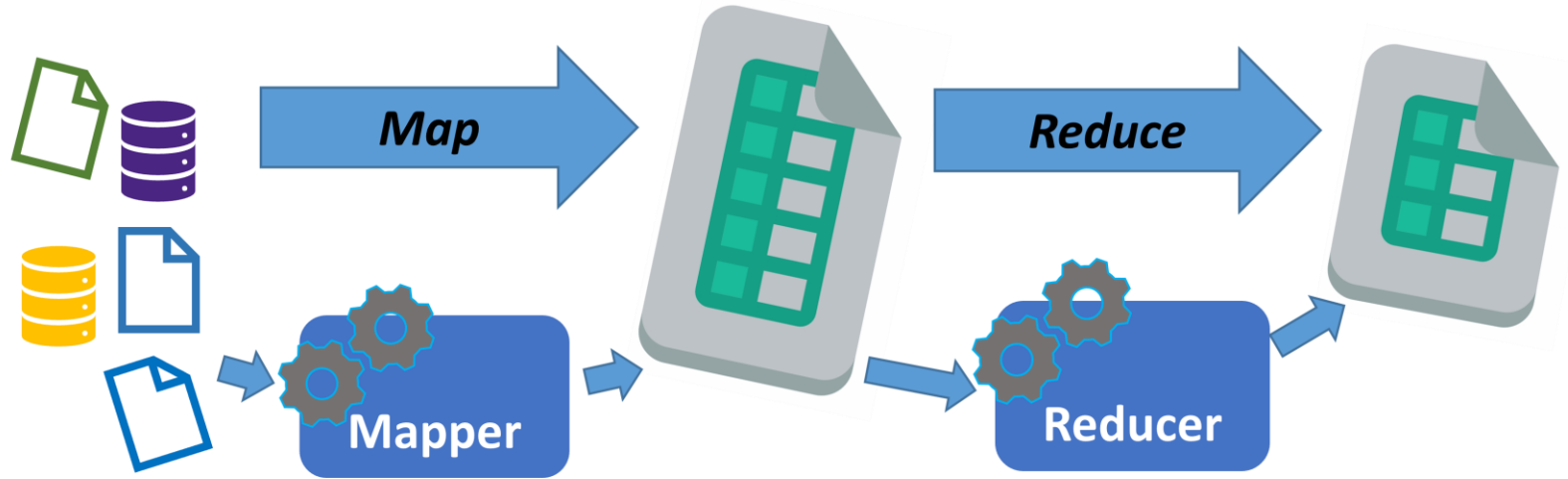
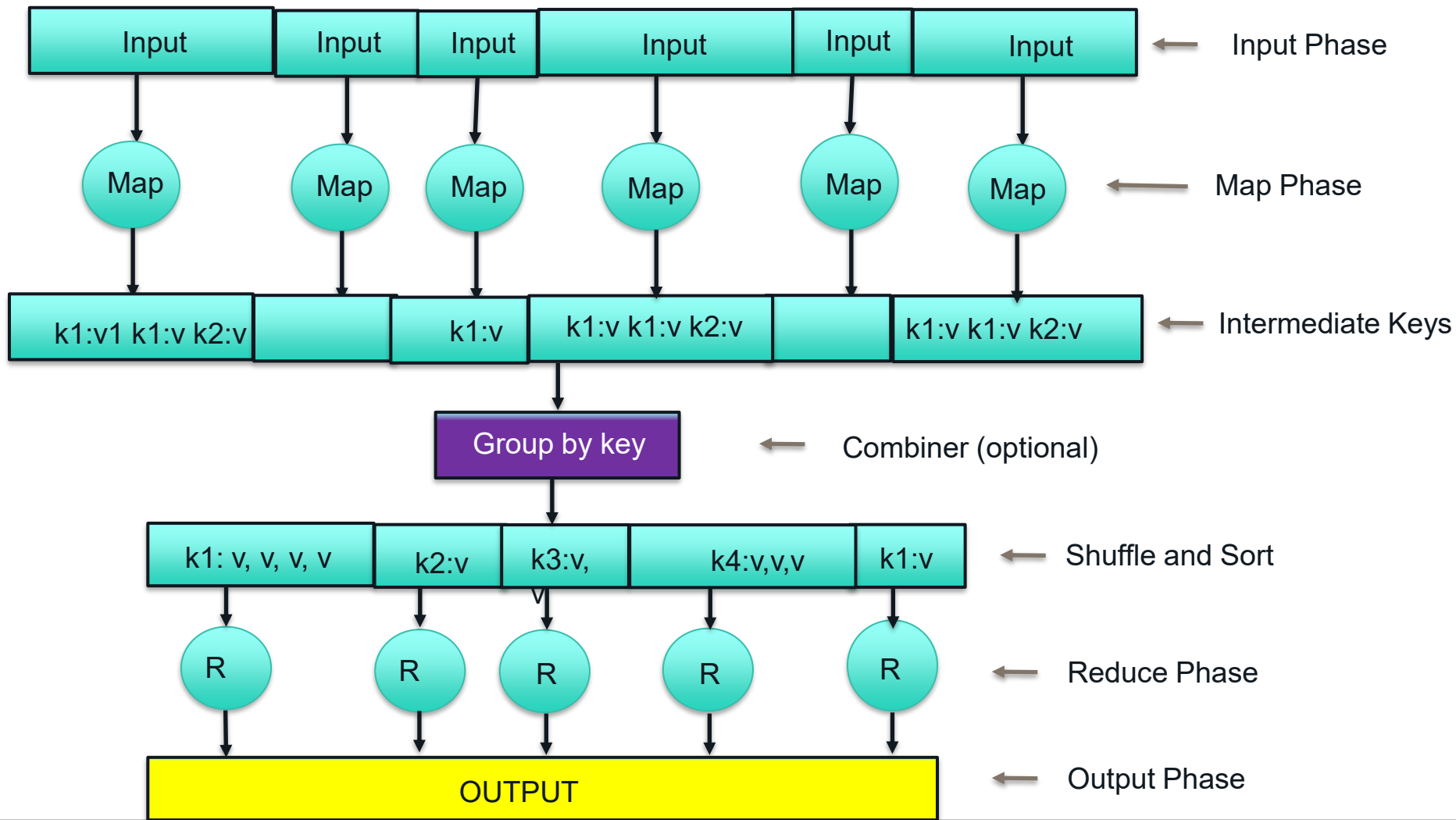


Image source: clipart-library.com

Phases in MapReduce Model





Input

Split

Map

Shuffle & Sort

Reduce

A B R
C C R
A C B

A B R

C C R

A C B

A, 1
B, 1
R, 1

C, 1
C, 1
R, 1

A, 1
C, 1
B, 1

A, 1
A, 1

B, 1
B, 1

C, 1
C, 1
C, 1

R, 1
R, 1

A, 2
B, 2
C, 3
R, 2

Hadoop Ecosystem (short)

Apache Hadoop

- Apache Hadoop is an open-source software utility that allows users to manage big data sets (from gigabytes to petabytes) by enabling a network of computers (or “nodes”) to solve vast and intricate data problems.
- **HDFS**: Hadoop Distributed File System

Hadoop MapReduce Ecosystem

- Hadoop Distributed File System (HDFS)
 - Primary data storage system that manages large data sets running on commodity hardware.
- Yet Another Resource Negotiator (YARN)
 - Cluster resource manager that schedules tasks and allocates resources (e.g., CPU and memory) to applications
- Hadoop MapReduce
 - Splits big data processing tasks into smaller ones, distributes the small tasks across different nodes, then runs each task
- Hadoop Common (Hadoop Core):
 - Set of common libraries and utilities that the other three modules depend on.

Hadoop MapReduce

› Mature and Statble

- › Hadoop MapReduce has been around for a long time and is considered stable and reliable

› Cost-effective

- › Designed to work efficiently on commodity hardware

› Fault Tolerance

- › HDFS provides fault tolerance by replicating data accross multiples nodes

Hadoop Use Cases

- Processing big data sets in environments where data size exceeds available memory
- Batch processing with tasks that exploit disk read and write operations
- Building data analysis infrastructure with a limited budget
- Completing jobs that are not time-sensitive
- Historical and archive data analysis

HDFS

- HDFS is implemented on the top of Linux file system

- HDFS Commands

`hdfs dfs -ls/put/get...`

References

1. Hadoop - MapReduce.
www.tutorialspoint.com/hadoop/hadoop_mapreduce.htm
2. MapReduce - Introduction.
https://www.tutorialspoint.com/map_reduce/map_reduce_introduction.htm